TIAGO FERRÃO-LINDQVIST

Test & Validation Engineer | Python, SQL & Test Data Automation

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PROFESSIONAL SUMMARY

Mechanical engineer with 12 years of experience in product design, testing and validation, transitioning into data engineering. Builds internal Python tools to process test-rig data, check results against specifications and produce engineering reports. Experience querying SQL Server, scheduling data processing on a lab PC and creating a Power BI dashboard for engineering teams.

TECHNICAL SKILLS

Data and programming: Python, pandas, NumPy, Matplotlib, npTDMS, SQL, SQL Server, SQLite, Power BI, Excel/VBA
Engineering and analysis: MATLAB/Simulink, ANSYS Mechanical, SolidWorks, CATIA V5, LabVIEW, Minitab, design of experiments, GD&T, DFMEA, 8D problem solving
Personal project technologies: Flask, Git, Docker Compose

PROFESSIONAL EXPERIENCE

Test & Validation Engineer
Corvane Automotive Components | Mar 2019–Present

- Developed Python scripts to parse TDMS and CSV output from six test rigs, check channels against test specifications and generate report figures, reducing report preparation from approximately three days to half a day per test.
- Scheduled the tool to run nightly on a lab PC using Windows Task Scheduler. It processes approximately 2 GB of data per week and writes summary results to an IT-created SQL Server staging table; the main test-results database is queried primarily read-only. Four other test engineers use the tool.
- Built a Power BI dashboard showing pass/fail trends and cycles to failure by product generation; approximately 15 validation and design engineers use it.
- Queried the plant's test-results database for trend analysis across three product generations. Database access is primarily read-only, with write access to the staging table.
- Plan and run durability, vibration and climatic tests on seat-adjuster mechanisms and write reports for OEM customers.
- Correlate rig results with ANSYS Mechanical FEA and lead 8D root-cause analysis of field failures. Proposed an adjuster rail bracket redesign that reduced warranty returns for that part by 40%.
- Maintain MATLAB/Simulink models of the adjuster drive to select rig load profiles matching customer duty cycles.
- Trained three apprentices in rig safety and test setup between 2022 and 2024.

Mechanical Design Engineer
Halvik Pumps | Sep 2014–Feb 2019

- Designed pump casings and impellers in SolidWorks, produced drawings with GD&T and owned DFMEA for two product lines.
- Conducted structural FEA in ANSYS and changed one casing family from grey cast iron to ductile iron, reducing material cost by 12%.

– Wrote a LabVIEW acquisition program for the pump performance test bench, reducing test setup time from approximately two hours to 30 minutes.

– Supported conversion of the test bench to a Siemens S7 PLC-controlled setup, adjusting setpoints and timers in TIA Portal under an automation engineer's supervision.

PERSONAL PROJECT

Bike-sharing Trips API | 2025

– Built a small Flask API with three read-only endpoints over a public bike-sharing dataset loaded into SQLite; runs locally on a laptop.

– Used Git and Docker Compose in a guided personal learning project.

EDUCATION AND TRAINING

MSc Mechanical Engineering, Sado Valley Polytechnic | 2014
Thesis: Fatigue life prediction of welded brackets under variable-amplitude loading

Online Data Engineering Course | In progress — 6 of 9 modules completed.

LANGUAGES

Portuguese: native | English: fluent | Spanish: good